8 4 2 1

1) 28 H →
0000 0000 0010 1000

a) entry → 5H
table → Global
privilege level → 00

b) Starting address → A3470000

Ending address →
(A3470000 + 0FFFFFFF)
= B346FFFF

c) A351BC12 = A3470000 + offset

offset = ABC12

8 4 2 1

d) Access right = 11 H

↳ 0001 0001

E = 0 , ED = 0 ⟶ data segment

e) A = 1 ∴ access granted

f) DPL = 00

2/a) 1023 — 5V

1 — $\frac{5}{1023}$ V

434 — $\frac{5 \times 434}{1023} = 2.12$ V

100 % — 4095

1 — $\frac{4095}{100}$

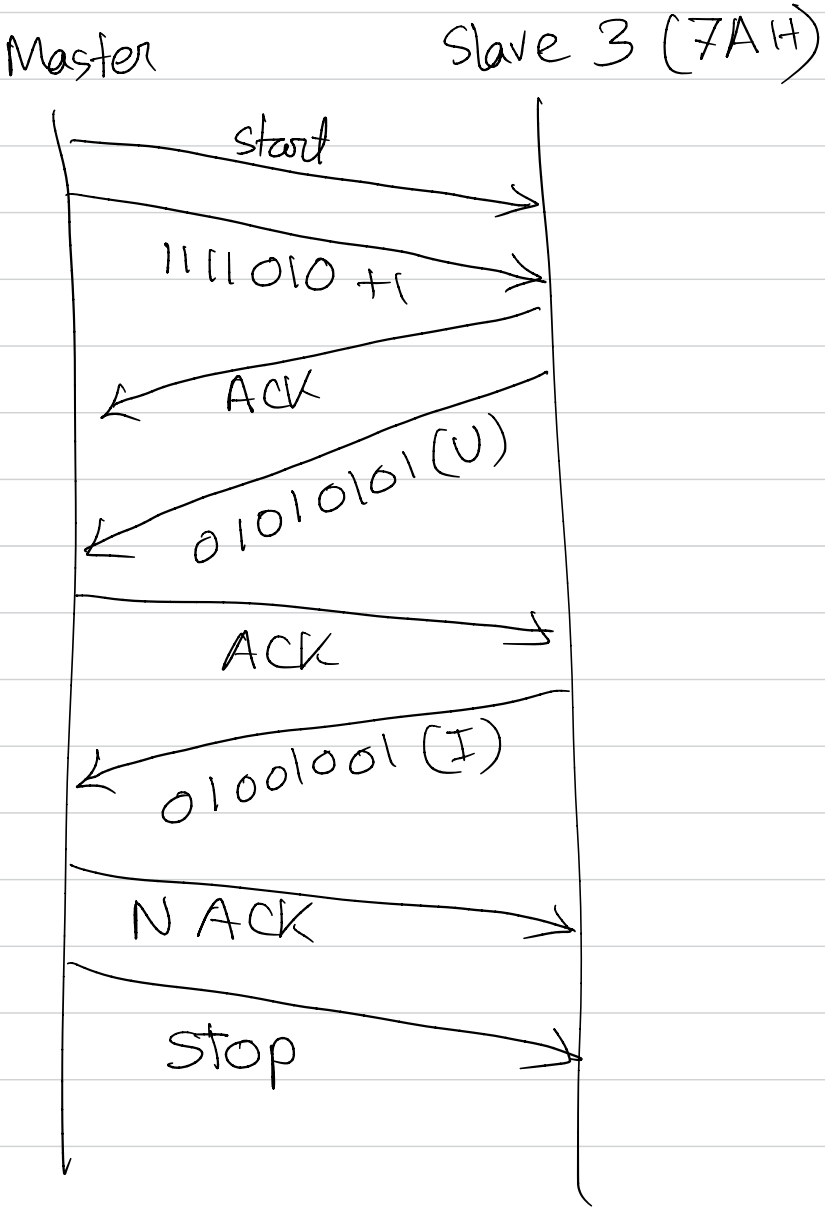
75 % — $\frac{4095 \times 75}{100}$

$\approx$ 3071 or 3072

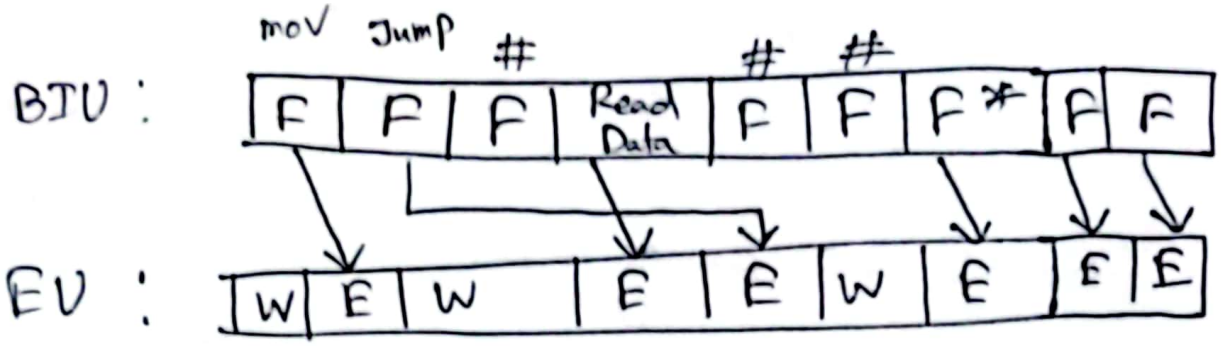
b) central monitoring unit → MP

sensor nodes → MC

b)


Master
Slave 3 (7AH)
start
1111010 +1
ACK
01010101 (U)
ACK
01001001 (I)
NACK
stop

3. a)



discard

b) i) Ready → 1 (Send Data) (MP ready)
→ 0 (Do not send data) (MP not ready)

ii) No, sending address → ALE = 1
" Data → $\overline{DEN}$ = 0

c) Paging is a memory segmentation technique used in V.M. Multitasking system.

4. a)
```
LDA   AH
SUB   CH
ADD   EH
SUB   FH
OUT
HLT
```

b)

Ins	AX
0H →	32/20H
1H →	22/16H
2H →	27/1BH
3H →	0/0H
9H →	0/0H
5H →	0/0H

c) 256×16 → RAM

256 → $2^n = 256$

$= 2^8$

$\therefore n = 8$ (A.B.)

16 → D.B. = 16 bit

Register		Size
P.C.	—	8 bit
MAR	—	8 "
Accumulator	—	16 "
B reg	—	16 "
Output "	—	16 "

CamScanner